



Hasnain Haroon

CIVIL ENGINEER & AI RESEARCHER

OPEN TO WORK & RESEARCH

CONTACT

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SKILLS & TOOLS

ENGINEERING SOFTWARE

AutoCAD ETABS Revit

Primavera P6 EPA SWMM

MS Excel MS Word Latex

AI & PROGRAMMING

Python YOLOv11 Computer Vision

Deep Learning Streamlit OpenCV

CORE COMPETENCIES

Structural Analysis RC Design

Steel Design ACI 318-19

Quantity Estimation Surveying

SOFT SKILLS

Leadership Technical Reporting

Project Management Problem Solving

Communication

LANGUAGES

Urdu NATIVE

English PROFESSIONAL

PORTFOLIO



hasnainharoonqureshi.github.io

PROFILE

Final-year Civil Engineering student at COMSATS University Islamabad (Wah Cantt), graduating September 2026 with a **3.75/4.00 CGPA**. Operates at the intersection of structural engineering and applied AI — developed a YOLOv11-based crack detection, segmentation and severity-classification platform deployed on both web and Android. Backed by four engineering internships spanning structural offices, commercial construction, residential sites and field surveying, with strong analytical, design and technical reporting capabilities.

3.75 CGPA / 4.00

4 Engineering Internships

Open House 2026

2026 Graduating Sep

EDUCATION

B.Sc. Civil Engineering

COMSATS University Islamabad, Wah Cantt Campus

Specializing in structural engineering with active research in AI-based structural health monitoring.

Treasurer, Institution of Civil Engineers (ICE) student chapter.

2022 – 2026

3.75 / 4.00

Structural Design

AI & CV Research

4 Internships

ICE Treasurer

Intermediate — FSc Pre-Engineering

Wah Hope College, Fateh Jang

Pre-Engineering stream with strong foundations in Mathematics, Physics & Chemistry.

2020 – 2022

FINAL YEAR PROJECT

Deep Learning-Based Structural Crack Detection, Segmentation, Width Estimation & Severity Classification

2nd Place — Open House & Job Fair 2026

YOLOv11s-seg · Sep 2025 – May 2026

AI system for automated structural health monitoring — detects, segments and classifies cracks in concrete (Hairline, Fine/Minor, Moderate, Severe), estimates crack width via pixel-to-mm calibration, and deploys across web (Streamlit) and Android platforms.

YOLOv11s-seg

Python

OpenCV

Streamlit

Android

Deep Learning

Live Demo →

EXPERIENCE

Trainee Engineer — Structure Office

Jul – Aug 2025

NESPAK (Pvt.) Ltd.

- Supported structural analysis and design documentation for multi-disciplinary infrastructure projects at Pakistan's largest engineering consultancy.
- Reviewed structural drawings, load calculations and design reports under senior engineer supervision.

Field Survey Camp

Aug 2024

COMSATS Wah — Harnoi, Abbottabad

- Intensive field training covering topographic survey, differential leveling, traversing and contouring across real terrain conditions.

Trainee Site Engineer

Jul – Aug 2024

Winston Developers (Pvt.) Ltd.

- Supervised construction of a multi-storey commercial building — quality control, material estimation and progress reporting.

Trainee Site Engineer

Jun – Jul 2024

Hashim Associates & Builders (Pvt.) Ltd.

- Residential construction supervision — reinforcement checking, formwork inspection, concrete pouring oversight and daily progress logs.

SELECTED PROJECTS

RC School Building Structural Design — ACI 318-14 · Jun 2025

Manual structural design of a 3-storey RC school building (120'×180') per ACI 318-14 & ASCE 7 — one-way slab, singly & doubly reinforced beams, column design with AutoCAD/Revit detailing.

ACI 318-14

ASCE 7

AutoCAD

Revit

Combined Sewerage System Design — EPA SWMM 5.2 · Dec 2025

Hydraulic simulation of a combined sewerage network in Wah Cantt — 168 subcatchments, dynamic wave routing and 11 outfalls with full continuity analysis.

EPA SWMM

Drainage Design

Hydrology

Residential Construction Planning & Scheduling — Primavera P6 · Jun 2025

Full project scheduling of a 10-Marla double-storey house using Primavera P6 — WBS, OBS, CPM, resource levelling and Earned Value Analysis (SPI 0.94, CPI 0.99) within a 12-month contract.

Primavera P6

CPM

Earned Value

Resource Levelling

ACHIEVEMENTS & CERTIFICATIONS

- 2nd Position — Open House & Job Fair Spring 2026, COMSATS Wah Campus (FYP competition)
- Treasurer, Institution of Civil Engineers (ICE) Student Chapter, CUI Wah (2025–2026)
- CPD Seminar — Finite Element Method for Structural Analysis (2025)
- Civil Nexus 2025 — Participation, ASCE University of Wah
- Certificate of Appreciation — Blood Donation Drive, Pakistan Red Crescent Society